LHNet: Laplacian Convolutional Block for Remote Sensing Image Scene Classification

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***Abstract*— Recently, many state-of-the-art results for remote sensing image scene classification have been achieved by convolutional neural networks (CNNs) due to their large learning capability. However, in the forward process of CNNs, the high-frequency/texture features are gradually blurred with hierarchical downsampling and convolution operations. High-frequency features are important to capture the diversity within a class and the similarity between classes. For example, the line features are crucial to distinguish a tennis court from a basketball court. For tennis court in different scenes, the highlight of line features can effectively avoid the influence of diverse backgrounds. As a consequence, we propose a Laplacian high-frequency convolutional block (LHCB) based on CNN to extract useful high-frequency features by trainable Laplacian operator. To propagate high-frequency features, we embed LHCB into the existing CNN structures and obtain LHNet. In LHNet, there are two pathways. The original CNN architecture can be taken as the low-frequency pathway, and we propose a high-frequency pathway (HFP) based on LHCB that propagates the residual high-frequency features blurred in each low-frequency layer. Considering that the high-frequency features usually show large variance between images of the same class, we propose a new objective for HFP to enhance the intraclass similarity of high-frequency features. The final objective function is obtained by combining the new objective and the baseline classification objective. Numerous experiments on three public available remote sensing image scene classification datasets, the Northwestern Polytechnical University-Remote Sensing Image Scene Classification (NWPU-RESISC45), Aerial Image Dataset (AID), and the University of California at Merced (UC Merced), demonstrate the superior performance of the proposed method.**

***Index Terms*— Convolutional neural networks (CNNs), Laplacian high-frequency convolutional block (LHCB), remote sensing image scene classification, representation learning.**

I. Introduction

IMAGE scene classification task is to classify images into a set of semantic classes according to the information inside the image [1]–[9]. As a fundamental problem in the field of computer vision, it has attracted more and more attentions in recent years [10]–[14]. From the feature extracting perspective, the existing image scene classification approaches can be roughly divided into three categories: handcrafted feature-based [15]–[18], [18]–[22], unsupervised feature learning-based [23]–[31], and deep learning-based approaches [32]–[39].

Early image classification approaches are usually based on handcrafted features, which are generated via extracting specific features, such as spectral, color, texture, and shape information or their combination by analyzing the images [15]–[18], [18]–[22]. Among them, color feature [20], texture feature [18], and histogram of oriented gradients (HOG)-based shape feature representation [15], [16] are most widely used. However, these approaches are difficult to extract rich semantic features, which are important behind a scene image, and thus, the classification performance is largely limited.

To overcome the limitations of the abovementioned handcrafted feature-based approaches, unsupervised feature-based approaches were then used for scene classification [23]–[31], [40]. They mainly focus on learning a set of basis functions used for feature encoding for an image. For early learning methods, the high-level features are generated from a series of handcrafted features to represent the main information via specific rules. For example, bag-of-visual words (BoVWs) [41], [42] usually generate features by performing unsupervised k-means clustering [43] on the set of handcrafted features or raw pixel intensity values. That is, these methods further process the handcrafted features to represent the abstract information of input images. Therefore, compared with handcrafted feature-based approaches, they can learn more discriminative features and are more suitable for representing the information of an image. Typical unsupervised feature learning methods

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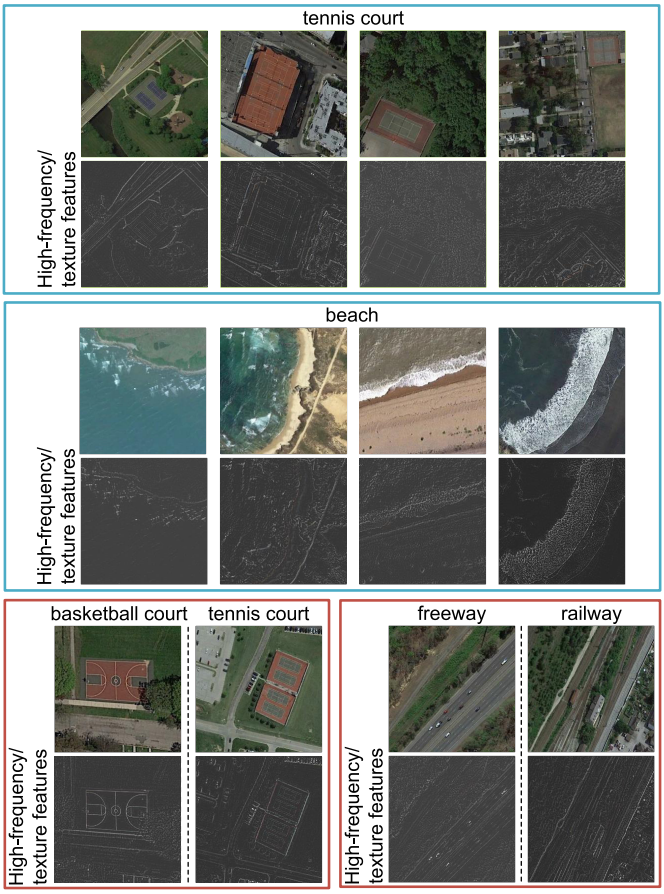


Fig. 1. Samples visualization of interclass similarity and intraclass diversity and the corresponding texture features (high-frequency features). The texture features are obtained by Laplacian decomposition. All these samples are taken from dataset NWPU-RESISC45 [1].

for scene classification include k-means clustering [43], sparse coding [23], [23], [25], [44], [45], autoencoder [24], and so on. The features in these approaches are learnable, which are sufficient and can adapt to input data. The performance of such approaches is usually better than that of handcrafted feature-based approaches. However, the discriminative performance of these approaches is still limited, as these approaches do not take use of class information and could not well represent distinguishable features between classes.

Nowadays, more and more deep learning-based learnable approaches were proposed, especially convolutional neural networks (CNNs)-based ones [14], [32]–[35]. These approaches can make use of class information to learn discriminative features from the original input data through CNN structure via optimizing an objective function of classification. Therefore, the features learned by CNNs are more robust for scene classification.

However, due to hierarchical filtering and downsampling processes of CNNs, high-frequency/texture features are gradually smoothed during the forward process [46]. Therefore, the traditional CNN architectures can be taken as the low-frequency pathway. For remote sensing image scene classification, high-frequency/texture features are crucial for identifying many objects, as shown in Fig. 1, where we show high-frequency features of some scenes. In Fig. 1, line features are important to distinguish tennis court and basketball court. Wave ripples are critical to identify scene of beach. To highlight and sufficiently extract the high-frequency features, we propose a Laplacian high-frequency convolutional block (LHCB) based on the CNN structure via trainable Laplacian operator. Laplacian operator could well extract the high-frequency information lost in the filtering and downsampling processes and has been widely used [47], [48]. To propagate high-frequency features, we propose a high-frequency pathway (HFP) via embed LHCBs into low-frequency CNN structures and construct LHNet. In addition, due to the variance of position, view angle, and size of objects in different images, the high-frequency features usually show large variance even within the same class. Therefore, we propose a new objective for the HFP considering intraclass similarity to better learn high-frequency features and further improve performance. Experimental results on three public available remote sensing image scene classification datasets demonstrate the effectiveness of the proposed LHCB, HFP, the object function, and the superiority performance of LHNet for remote sensing image scene classification.

In summary, the core contributions of this work can be summarized as follows.

1) We propose an LHCB to effectively extract high-frequency features blurred in CNN forward process via trainable Laplacian operator. The high-frequency features are particularly important for remote sensing image scene classification, as shown in Fig. 1.
2) To well propagate the high-frequency features, we construct LHNet by embedding LHCB into exact locations in a low-frequency CNN structure. In this way, LHNet could adaptively propagate the appropriate low- and high-frequency feature representation of image scenes.
3) To well learn robust high-frequency feature representation, we further propose a new object function considering intraclass similarity for HFP.
4) Experimental results demonstrate the effectiveness of the proposed LHCB, HFP, and the object function. It is also demonstrated that HFP could be constructed based on many existing networks to further improve the performance and achieve state-of-the-art performance.

The rest of this article is organized as follows. Section II introduces image scene classification methods closely related to the method proposed in this article. Section III introduces the proposed LHNet in terms of architecture, objective function, and corresponding analysis. Section IV presents the experimental setup and numerous experimental results in detail. Finally, Section V summarizes this article.

II. Related Work

In this section, we introduce three types of methods closely related to the proposed method, namely, CNN-based remote sensing image scene classification methods, texture feature representation in CNNs, and Laplacian decomposition-based methods for texture feature extraction.

A. CNNs for Image Scene Classification

The architecture of a CNN, which determines the information flow, is crucial for the performance of it. Many

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improvements have been proposed for better representation of input images [32]–[39]. Visual geometry group (VGG) network [32] was proposed by studying the relationship between network depth and accuracy in large-scale image recognition problems. Residual network (ResNet [33]) was proposed by introducing identity mapping on the basis of traditional CNN structure. Inception network (module in GoogLeNet [34]) was proposed by increasing the depth and width of the network while keeping the computational budget constant. Squeeze-and-excitation network (SENet [35]) was proposed by introducing an SE block to adaptively recalibrate channel-wise feature responses by explicitly modeling interdependencies between channels. A convolutional block attention module (CBAM [37]) was proposed by combining both channel attention (CA) module and spatial attention (SA) module.

Remote sensing images show large difference compared with general images. Nowadays, with the development of CNNs, many excellent approaches based on CNNs have been proposed for remote sensing image scene classification [3], [14], [49]–[57]. For example, discriminative-CNN (D-CNN [14]) was proposed by simultaneously optimizing metric learning regularization (contrastive embedding) and cross entropy through the current CNN structure. Due to the introduction of metric learning, in the D-CNN's feature space, the feature distance between the same classes is small, and that between different classes is large. Thus, the performance of D-CNN was significantly improved compared with their baselines. Multigranularity canonical appearance pooling (MG-CAP [49]) was proposed via automatically capturing the latent ontological structure of remote sensing datasets to improve performance. A method in [50] investigated adversarial training (Adv.Tr.) strategy to resolve the problem existed in most of the state-of-the-art deep neural networks that adversarial examples may be misclassified into the wrong category. Perturbation-seeking generative adversarial networks (PSGANs [3]) were proposed by designing a new training framework and developing a reconstruction method to improve the defensive ability for unknown attacks. Skip-connected covariance (SCCov) network [51] was proposed by embedding novel skip connections and covariance pooling into the traditional CNNs to achieve more representative feature learning.

B. Texture Feature Representation in CNNs

Since texture features are important in many practical applications, many approaches have been proposed to mine texture feature representations of objects [58]–[61] in CNNs. Considering that the overall shape information extracted by the fully connected layers of a classic CNN is of minor importance in texture analysis, a method in [58] pooled an energy measure from the last convolution layer to discard the overall shape information and focused on texture features. A texture descriptor was proposed in [59], which is obtained by fisher vector pooling of a CNN filter bank. Wavelet and contourlet transforms are used in [64] and [65] to enhance the texture features of CNN structures. For scene classification, TEX-Net was proposed in [62] where the network is trained using mapped coded images with explicit local binary patterns (LBPs)-based texture information that provides complementary information. A frequency-domain branch was introduced in CNN for scene classification in [63] with Fourier transform. In these methods, texture features are extracted by the existing transforming templates, such as Fourier and wavelet transforms, and, therefore, obtained good performance in specific tasks.

Different from texture feature extraction methods mentioned earlier, in this article, to adaptively extract texture features in remote sensing image scene classification, we propose an LHCB based on CNNs via trainable Laplacian operator.

C. Laplacian Operator for Texture Feature Extraction

The images in Laplacian pyramid generated by Laplacian decomposition are a series of differential images containing high-frequency information of an object [64]. For an input image I, the Laplacian decomposition (Lapdec) process can be described as follows:

$$\left(\mathrm{Lap}_2^I, \mathrm{Lap}_3^I, \ldots, \mathrm{Lap}_l^I, \ldots, \mathrm{Lap}_L^I\right) = \mathrm{Lapdec}_L(I) \tag{1}$$

where L donates decomposition levels. Each image in the lth pyramid level (Lap_l^I) could be calculated by the following formula:

$$\mathrm{Lap}_l^I(i,j) = \mathrm{Gau}_l^I(i,j) - \omega_{m\times n}^{\mathrm{Gau}}\left(\left(\mathrm{Gau}_{l+1}^I(\lfloor i/2\rfloor, \lfloor j/2\rfloor)\right)\right) \tag{2}$$

where $\omega_{m\times n}^{\mathrm{Gau}}$ donates the Gaussian filter of size $m \times n$, and Gau_l^I represents Gaussian image of the l level in Gaussian pyramid corresponding to image I

$$\mathrm{Gau}_l^I(i,j) = \omega_{m\times n}^{\mathrm{Gau}}\left(\mathrm{Gau}_{l-1}^I(2i, 2j)\right) \tag{3}$$

where $\mathrm{Gau}_1^I = I$ and (i, j) denotes the coordinates position of a pixel. In general, Laplacian pyramid is computed by subtraction operator on each level. Then, the residual high-frequency information between adjacent levels of Gaussian pyramid can be extracted.

Since the Laplacian feature pyramid generated by Laplacian operator could well represent the high-frequency/texture features of the object, the Laplacian operator has been widely used in image super-resolution [47], [48], image blending [64], texture synthesis [65], edge-aware filtering [66], semantic segmentation [67], object detection [46], and so on.

As analyzed in Section I, it is crucial to extract high-frequency/texture features to improve the classification performance of remote sensing image scenes. Traditional CNN models with hierarchical filtering and downsampling operators can be taken as the construction of Gaussian pyramid (here, we take it as low-frequency pathway). Inspired by the Laplacian operator, we propose an LHCB via trainable Laplacian operator between two convolutional blocks in the CNN structure. Based on LHCB, we propose an HFP corresponding to the low-frequency path of CNN. Both low-frequency pathway and HFP construct the proposed LHNet. Thus, LHNet could learn the suitable low- and high-frequency features of the scene through low-frequency pathway and HFP. To further improve performance, considering that the feature variance of high-frequency features

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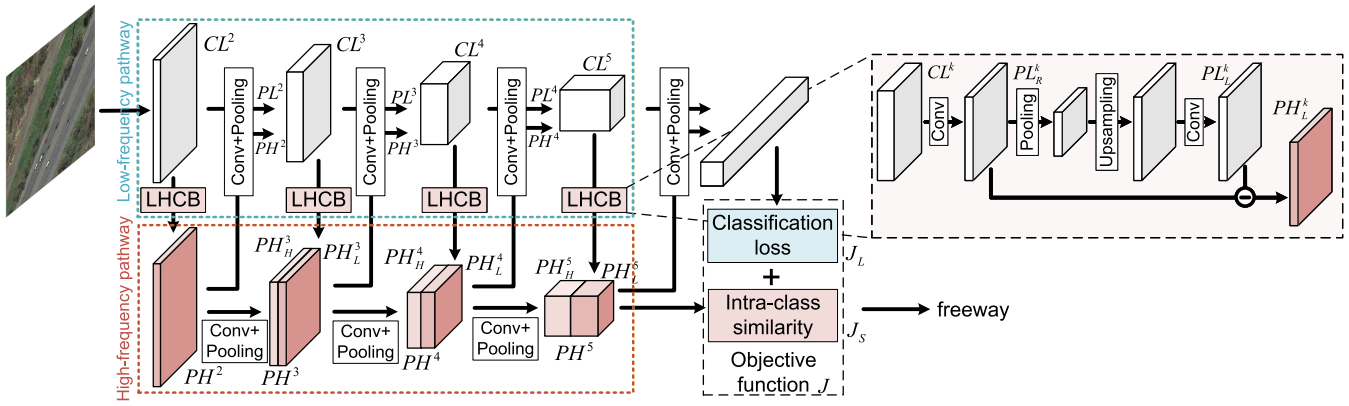


Fig. 2. Scheme of the proposed LHNet.

between images is usually large even within the same class, we design a new objective function to enhance intraclass similarity of high-frequency features. Moreover, LHNet could be embedded into many existing CNN structures to further improve the performance.

III. Proposed LHNet

Due to the hierarchical convolution and pooling operators, high-frequency/texture features are usually smoothed during the forward computation of CNNs. As introduced earlier, high-frequency information is crucial to recognize many objects in remote sensing images. Therefore, we propose a novel LHNet based on CNN to extract, propagate, and learn both low- and high-frequency features of the object. In the following, we describe the LHNet from the architecture, objective function, optimization, constructions on different baselines, and corresponding analysis.

A. Architecture

Fig. 2 shows the architecture of the LHNet. LHNet is composed of low-frequency pathway, HFP, and corresponding connections. The two pathways interact with each other during the construction of them. In CNN, with hierarchical convolutional and pooling operators, the semantic features can be extracted in the top layer. However, convolutional operators are usually difficult to deal with high-frequency features. Therefore, we take the forward process of CNN as the low-frequency pathway and construct an HFP to extract high-frequency features. The HFP is constructed by LHCB via trainable Laplacian operators on low-frequency convolutional blocks.

1) Low-Frequency Pathway: Following CNN, the feature maps CL^k in the kth convolutional layer are generated via convolutional operator:

$$CL^k = \mathrm{Conv}^L_{m\times n}\left(\left[PL^{k-1}, PH^{k-1}\right]\right) \tag{4}$$

where $PL^{k-1} \in \mathbb{R}^{H\times W\times C_k}$, in which H, W, and C_k are the height, width, and channel of it, which is a 3-D tensor denoting the pooled feature maps in the $(k-1)$th pooling layer in low-frequency pathway. Symbol [ ] donates *Concat* operator. PH^{k-1} denotes the feature maps in the $(k-1)$th pooling layer in HFP. Then, the feature maps PL^k in the kth pooling layer are obtained by pooling operator from CL^k. After hierarchical convolutional and pooling operators, the output of the network is obtained via fully connecting layers, and the objective function for classification is defined.

2) High-Frequency Pathway: HFP mainly generates hierarchical high-frequency feature maps via trainable Laplacian operators from feature maps of low-frequency pathway. It follows the basic architecture of CNN, which is also composed of hierarchical convolutional and pooling layers. The feature map set PH^k in the kth pooling layer is computed from two parts including those propagated from lower layers PH^{k-1} via convolutional and pooling operators (denoted by PH^k_H) and those generated via LHCB from feature maps of corresponding low-frequency layer CL^k (denoted by PH^k_L).

The architecture of LHCB is illustrated in Fig. 2. First, a convolutional layer is used to regularize CL^k into a tensor $PL^k_R \in \mathbb{R}^{H\times W\times C_k/2}$ with half channels

$$PL^k_R = \mathrm{Conv}_{1\times 1}\left(CL^k\right). \tag{5}$$

Then, each feature map in PL^k_R is downsampled to generate low-level feature map and then upsampled. A convolutional layer follows the upsampled feature map to generate a tensor PL^k_L, which contains low-frequency information of PL^k_R:

$$PL^k_L(i,j) = \mathrm{Conv}_{m\times n}\left(2\times up\left(\mathrm{Pool}\left(PL^k_R\right)\right)\right) \tag{6}$$

where (i, j) denotes the position of each pixel, and $2\times$ up denotes the $2\times$ neighborhood upsampling operator, which upsamples the spatial resolution by a factor of 2 using nearest neighborhood upsampling as in [68]. Finally, the high-frequency features can be extracted via subtraction operators

$$PH^k_L = PL^k_R - PL^k_L. \tag{7}$$

The whole process is equivalent to Laplacian operators with trainable filtering kernels.

$PH^k_H \in \mathbb{R}^{H\times W\times C_k/2}$ is a tensor generated from lower layer of HFP following the forward process of CNN. Then, the feature maps of the kth pooling layer of HFP, i.e., PH^k, are generated via Concat operator

$$PH^k = \left[PH^k_H, PH^k_L\right]. \tag{8}$$

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Finally, as shown in Fig. 2, PH^k is fed into the $k+1$th layer in both low-frequency pathway and HFP. After the last pooling layer, both low- and high-frequency feature maps are fed into fully connecting layers to generate the output of the whole network where the objective function for classification is defined.

Different from the Gaussian filter assigned in Laplacian pyramid in [64], the filter in the construction of LHCB is trainable by optimizing an objective. The trainable filter could make the network learn more adaptively low- and high-frequency features of the scene and, therefore, improves the performance. To learn more discriminative high-frequency features, except for classification loss, we define another objective especially for the HFP.

B. Objective Function

For classification, a loss function is defined as the loss between output of the network and label. Many loss functions can be used according to different baselines. Here, we denote the loss function with J_L (here, softmax is used). However, for high-frequency features, the variance between images is usually large, even within the same class. As shown in Fig. 1, due to the variance of position, view angle, and size of objects in different images, the detailed information usually shows distinct features within corresponding pixels. Since part of the high-frequency features is generated via Laplacian operator from low-frequency features, the final features will influence the discriminative capability of the network due to weak intraclass similarity of high-frequency features. Therefore, we define another objective to enhance the intraclass similarity of high-frequency features

$$\min J_S = \min \mathrm{Var}(f_H) = \sum_c \frac{\sum_n \left\| f_H\left(x_n^c\right) - \mu_c \right\|_2^2}{N_c} \tag{9}$$

where $f_H(x_n^c)$ denotes the output feature in the HFP of the data that belong to the cth class. μ_c is the mean feature of the cth class. N_c is the number of samples in the cth class. J_S means to reduce the variance of high-frequency features within the same class. The final objective is obtained by combining the two terms

$$\min J = \min J_L + \lambda J_S \tag{10}$$

where λ is a user-defined parameter that controls the importance of the two terms. To optimize the objective function, it is also important to derive the backpropagation pathways to compute the gradient of parameters in each layer.

C. Optimization

We use backpropagation to optimize the model, and the forward process follows the architecture described earlier. It is crucial to derive the gradient of the last layers, respectively, of low-frequency pathway and HFP. That of low-frequency pathway follows the gradient of objective J_L, which is widely discussed in many works. In J_S, there are two types of parameters, including network parameters and μ_c. Here, we first optimize the network parameters and then update μ_c. Then, for each data x_n^c, the gradient of the output layer of HFP δ_H^l is derived as

$$\delta_H^l = \frac{\partial J_L}{\partial PH^l} + \lambda \frac{\partial J_S}{\partial PH^l} = \mathrm{back}\left(\delta_L^{l+1}\right) + \lambda\left(PH^l - \mu_c\right) \tag{11}$$

where l denotes the last layer of the HFP. δ_L^{l+1} denotes the gradient of the $l+1$th layer in the low-frequency pathway, and back$(\cdot)$ denotes the backpropagation operator of CNN. Then, for the kth layer in HFP, the gradient is backpropagated from higher layers in both low-frequency pathway and HFP: $\delta_H^k = \mathrm{back}(\delta_H^{k+1}) + \mathrm{back}(\delta_L^{k+1})$.

For low-frequency layers, such as PL^k, the gradient of them contains not only that backpropagated from higher low-frequency layer but also from part of corresponding high-frequency layer δ_H^k through the backward of LHCB. In LHCB, $-\delta_H^k$ first backpropagates to the layer of PL_R^k with gradient denoted by δ_R^k. Then, $\delta_R^k = \delta_R^k + \delta_H^k$, which is the gradient of trainable Laplacian operator. δ_R^k backpropagates through one convolutional layer to generate the gradient in high-frequency layer fed to the corresponding low-frequency layer. With gradient of each layer, the connecting weights can be updated. For details, we use the Adam optimizer [69] to update the network parameters.

D. Effectiveness Analysis

As the CNN goes forward, the high-frequency features of the previous layer in the CNN are blurred by convolution and pooling. The Laplacian operation upsamples the feature map of the latter layer and then subtracts the feature map of the previous layer to obtain this high-frequency feature map, which can restore the blurred high-frequency features to a certain extent. This process is beneficial to remote sensing image classification tasks. In addition, in our proposed method, the extraction of high-frequency features adopts the convolution operation; that is, the acquisition of this high-frequency feature is a trainable process. Therefore, through learning, our proposed method can adaptively learn appropriate high- and low-frequency feature representations.

To intuitively demonstrate the effectiveness of high-frequency features, we exhibit the low- and high-frequency feature maps in the second, third, and fourth layers with input of two similar scenes in Fig. 3. As shown in Fig. 3, baseball court and basketball court show similar appearance, and both shape and color of them are the same. The only feature that can distinguish them is the textures within them. However, in the low-frequency pathway, the textures cannot be well retained with hierarchical filtering and downsampling operators. Therefore, some low-frequency features show similar features of the two scenes, especially in the higher layers. HFP specially highlights the texture information. Therefore, the high-frequency feature maps show more different features of the two scenes.

E. Constructing LHNet on Different Baselines

Except for basic CNN, as many network blocks, LHNet can also base on different baselines, such as SENet [35] and

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TABLE I
ResNet50 [33], LH-ResNet50, SENet50 [35], AND LH-SENet50 WITH A $32 \times 4d$ TEMPLATE

Output size	ResNet50	LH-ResNet50	SENet50	LH-SENet50
112×112	conv, 7×7, 64, stride 2			
	max pool, 3×3, stride 2			
56×56	$\begin{bmatrix} conv, 1 \times 1, 64 \\ conv, 3 \times 3, 64 \\ conv, 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} conv, 1 \times 1, 64 \\ conv, 3 \times 3, 64 \\ conv, 1 \times 1, 256 \end{bmatrix} \times 3$ $\begin{bmatrix} conv, 1 \times 1, 128 \\ conv, 3 \times 3, 128 \end{bmatrix}$	$\begin{bmatrix} conv, 1 \times 1, 64 \\ conv, 3 \times 3, 64 \\ conv, 1 \times 1, 256 \\ fc, [16, 256] \end{bmatrix} \times 3$	$\begin{bmatrix} conv, 1 \times 1, 64 \\ conv, 3 \times 3, 64 \\ conv, 1 \times 1, 256 \\ fc, [16, 256] \end{bmatrix} \times 3$ $\begin{bmatrix} conv, 1 \times 1, 128 \\ conv, 3 \times 3, 128 \end{bmatrix}$
28×28	$\begin{bmatrix} conv, 1 \times 1, 128 \\ conv, 3 \times 3, 128 \\ conv, 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} conv, 1 \times 1, 128 \\ conv, 3 \times 3, 128 \\ conv, 1 \times 1, 512 \end{bmatrix} \times 4$ $\begin{bmatrix} conv, 1 \times 1, 256 \\ conv, 3 \times 3, 256 \end{bmatrix}$	$\begin{bmatrix} conv, 1 \times 1, 128 \\ conv, 3 \times 3, 128 \\ conv, 1 \times 1, 512 \\ fc, [32, 512] \end{bmatrix} \times 4$	$\begin{bmatrix} conv, 1 \times 1, 128 \\ conv, 3 \times 3, 128 \\ conv, 1 \times 1, 512 \\ fc, [32, 512] \end{bmatrix} \times 4$ $\begin{bmatrix} conv, 1 \times 1, 256 \\ conv, 3 \times 3, 256 \end{bmatrix}$
14×14	$\begin{bmatrix} conv, 1 \times 1, 256 \\ conv, 3 \times 3, 256 \\ conv, 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} conv, 1 \times 1, 256 \\ conv, 3 \times 3, 256 \\ conv, 1 \times 1, 1024 \end{bmatrix} \times 6$ $\begin{bmatrix} conv, 1 \times 1, 512 \\ conv, 3 \times 3, 512 \end{bmatrix}$	$\begin{bmatrix} conv, 1 \times 1, 256 \\ conv, 3 \times 3, 256 \\ conv, 1 \times 1, 1024 \\ fc, [64, 1024] \end{bmatrix} \times 6$	$\begin{bmatrix} conv, 1 \times 1, 256 \\ conv, 3 \times 3, 256 \\ conv, 1 \times 1, 1024 \\ fc, [64, 1024] \end{bmatrix} \times 6$ $\begin{bmatrix} conv, 1 \times 1, 512 \\ conv, 3 \times 3, 512 \end{bmatrix}$
7×7	$\begin{bmatrix} conv, 1 \times 1, 512 \\ conv, 3 \times 3, 512 \\ conv, 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} conv, 1 \times 1, 512 \\ conv, 3 \times 3, 512 \\ conv, 1 \times 1, 2048 \end{bmatrix} \times 3$ $\begin{bmatrix} conv, 1 \times 1, 1024 \\ conv, 3 \times 3, 1024 \end{bmatrix}$	$\begin{bmatrix} conv, 1 \times 1, 512 \\ conv, 3 \times 3, 512 \\ conv, 1 \times 1, 2048 \\ fc, [128, 2048] \end{bmatrix} \times 3$	$\begin{bmatrix} conv, 1 \times 1, 512 \\ conv, 3 \times 3, 512 \\ conv, 1 \times 1, 2048 \\ fc, [128, 2048] \end{bmatrix} \times 3$ $\begin{bmatrix} conv, 1 \times 1, 1024 \\ conv, 3 \times 3, 1024 \end{bmatrix}$
1×1	global average pool, C-d fc, softmax			

ResNet [33]. To construct the LHNet, an HFP generated by LHCB is integrated into the baseline network with each layer corresponding to those in baseline network. To illustrate this point, we construct LHNet by absorbing the HFP into two complicated structures, as shown in Fig. 4.

As illustrated in Fig. 4, the LHNet can be constructed directly based on the ResNet [33] and SENet [35]. In Fig. 4, the LHNet based on ResNet is denoted as Laplacian high-frequency-residual network (LH-ResNet) and that based on SENet as Laplacian high-frequency-sequeeze-and-excitation network (LH-SENet). For LH-ResNet, we also use a residual connection for each block in the HFP. SENet improves ResNet via an SE block. It is defined to highlight useful channels. Since the SE block is also additionally embedded into ResNet, to reduce computational complexity, we construct the HFP as in LH-ResNet. Table I lists a detailed and clear comparison between ResNet50 [33] and LH-ResNet50, as well as SENet50 [35] and LH-SENet50.

IV. EXPERIMENTS

A. Datasets Description and Experimental Setup

To verify the performance of the proposed method, we conduct experiments on three public available remote sensing image scene datasets. These datasets are NWPU-RESISC45 [1], AID [11], and UC Merced [70]. Scenes with obvious texture features are extensively existed in these three datasets. The details of these three datasets are listed in Table II.

In our experiments, the divisions of the training sets and testing sets for these datasets are also shown in Table II. The divisions are the same to those in [1], [11], [14], and [74] for fair comparison. To obtain more reliable results, we conduct ten cross-validation experiments at each training ratio (T.R.; training samples are randomly selected from the dataset under each T.R). Finally, the average and standard deviation of the classification results of these ten runs are listed. All experiments are run on the same workstation (Intel Xeon CPU 643 E5-2630 v3 2.4 GHz $\times$ 32, 64-GB RAM) with the platform of Caffe 2 on operating system of Ubuntu 16.04.

We chose ResNet [33] (ResNet50 and ResNet101) and SENet [35] (SENet50 and SENet101) as the backbone networks. These backbone networks are pretrained on Imagenet and then fine-tuned on the target datasets. A total of 15 000 iterations are trained. The learning rate is 0.01 and 10% declined after 1000 iterations. λ is set to 0.1 in all experiments.

B. Evaluation Metrics

We use the overall accuracy (OA) and confusion matrix to quantify the classification results as in [14]. The OA reflects the percentage of correctly classified images in total test images. In this case, the class information is not considered. The confusion matrix is a table that reflects errors and confusions among classes. In particular, the confusion matrix accumulates the number of correctly classified and misclassified images for each class and reflects them as a percentage in the table.

C. Evaluation on LHCB

To verify the effectiveness of the proposed LHCB, we conduct experiments on three datasets, NWPU-RESISC45 [1],

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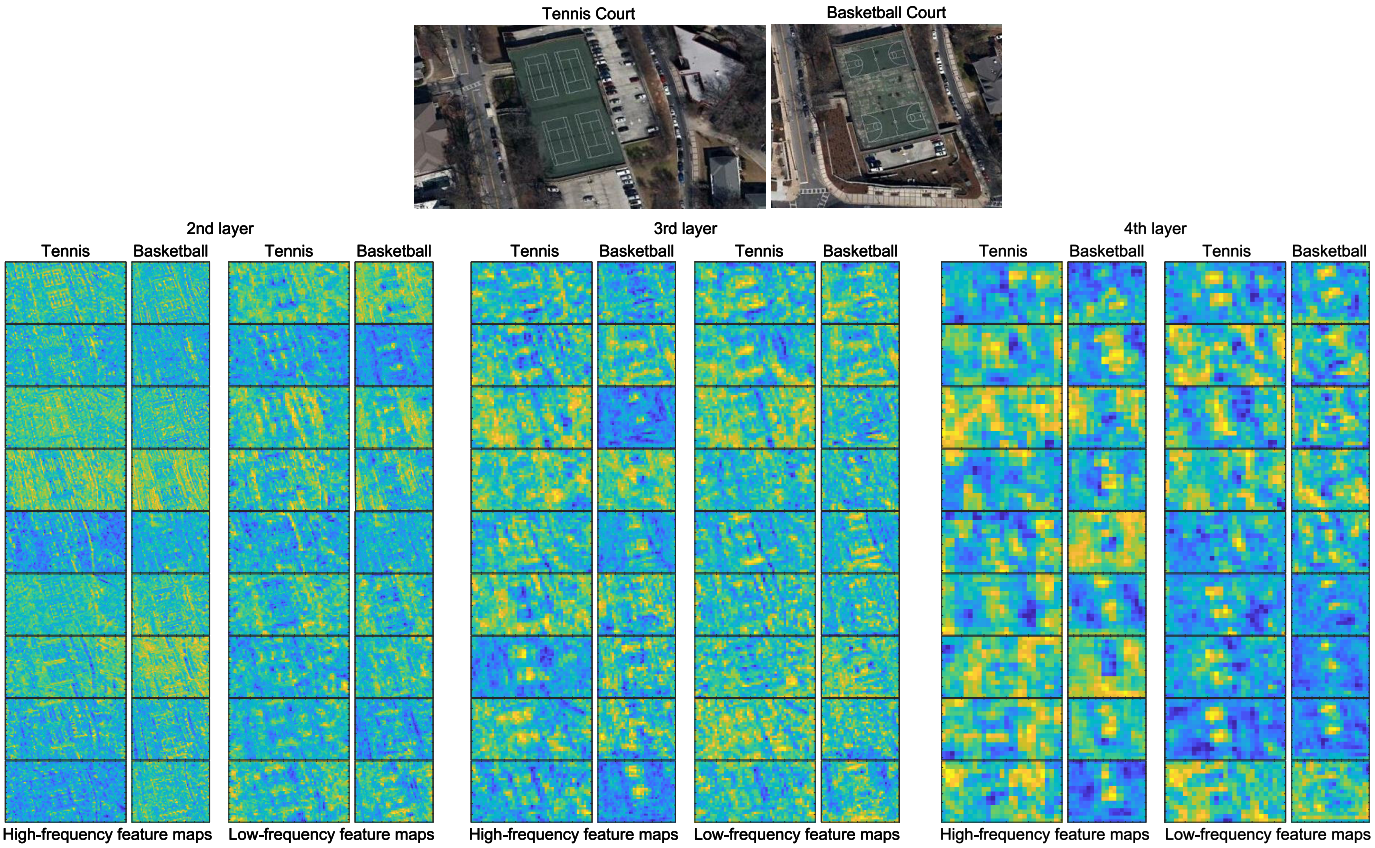


Fig. 3. Comparison of low- and high-frequency feature maps on two similar scenes. Two similar scenes of basketball court and tennis court are shown at the top of the figure followed by feature maps generated in different layers of HFP and low-frequency pathway on the two scenes. We choose nine typical feature maps in different layers with input of tennis and basketball courts, respectively. Each column denotes the nine feature maps in one layer. By comparison, some low-frequency feature maps show similar appearance with different input scenes, while the difference can be well distinguished in high-frequency feature maps.

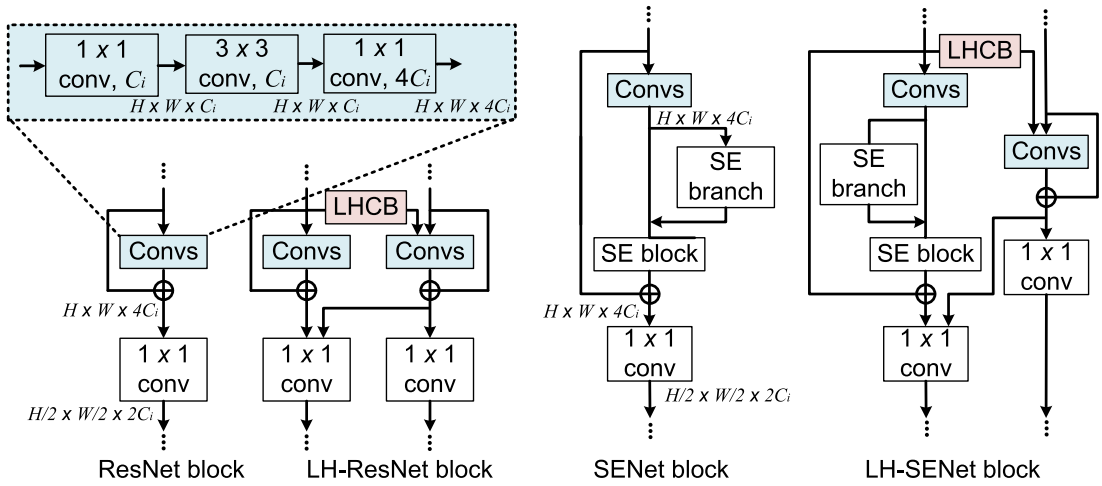


Fig. 4. Schemas of the original ResNet and SENet modules and the corresponding LH-ResNet and LH-SENet modules.

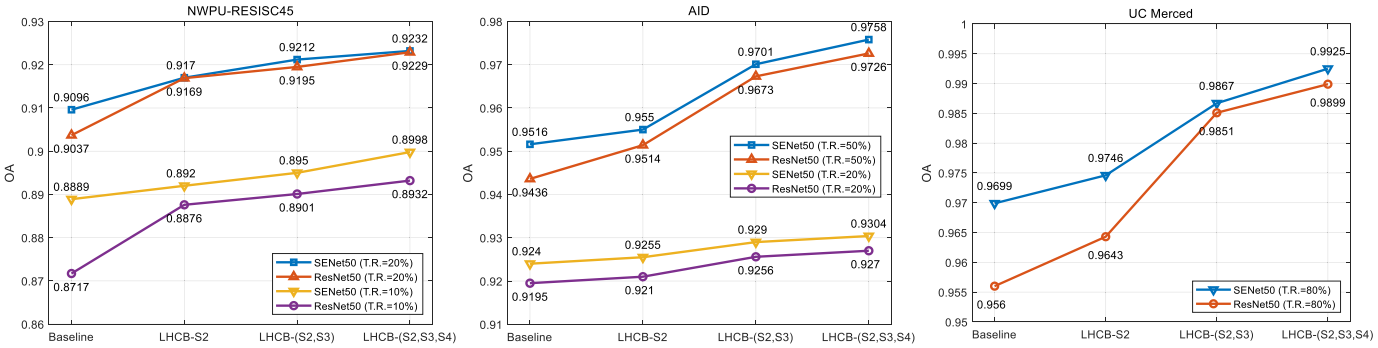
AID [11], and UC Merced [70]. We choose the two classic neural networks, including ResNet [33] and SENet [35], as the baselines. We first verify the performance impact of adding LHCB to different stages (stage_2, stage_3, and stage_4). The original definition of *stage* in residual block can be found in [33]. Different downsample methods can be used in LHCB, so we then verify the performance impact of different downsample methods in LHCB. Convolutional kernel $\text{Conv}_{m\times n}$ is also a variable that affects the performance of LHCB. Finally, we conduct experiments to verify the effect of the size of $\text{Conv}_{m\times n}$ on LHCB.

1) Impact on Different Stages: Fig. 5(a)–(c) shows the quantify comparison results of ResNet with LHCBs in different stages (stage_2, $\text{stage}_{2,3}$, and $\text{stage}_{2,3,4}$) and SENet with LHCBs in different stages (stage_2, $\text{stage}_{2,3}$, and $\text{stage}_{2,3,4}$) on datasets, NWPU-RESISC45, AID, and UC Merced, respectively. From

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TABLE II
DESCRIPTION OF DATASETS NWPU-RESISC45 [1], AID [11], AND UC MERCED [70]

NWPU-RESISC45 [1]	AID [11]	UC Merced [70]
Total images: 31500 Total classes: 45 Images per class: 700 Image size: 256×256 Spatial resolution: 0.2-3m Scene classes: Airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland.	Total images: 10000 Total classes: 30 Image per class: 220-420 Image size: 600×600 Spatial resolution: 0.5-8m Scene classes: Airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct.	Total images: 2100 Total classes: 21 Images per class: 100 Image size: 256×256 Spatial resolution: 0.3m Scene classes: Agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tank, and tennis court.
Training ratio (T.R.) = 10%/20%	T.R. = 20%/50%	T.R. = 80%



(a) (b) (c)

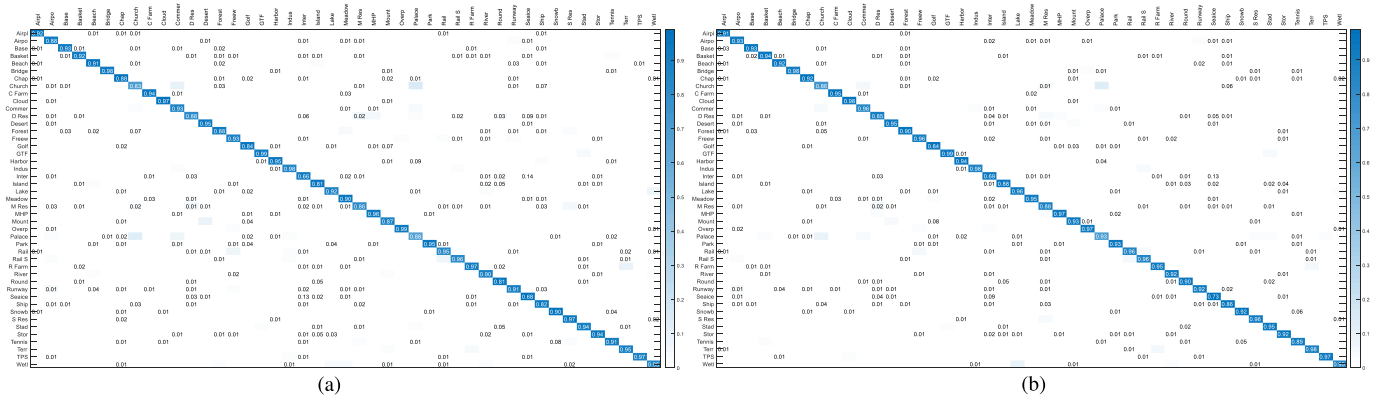
Fig. 5. Effect of integrating LHCB with ResNet50 [33] and SENet50 [35] at different stages on (a) NWPU-RESISC45 [1], (b) AID [11], and (c) UC Merced [70] datasets. S2, S3, and S4 indicate stage_2, stage_3, and stage_4 in turn.

experimental results on dataset NWPU-RESISC45 shown in Fig. 5(a), it can be found that the ResNet50 with LHCBs on stage_2, $\text{stage}_{2,3}$, and $\text{stage}_{2,3,4}$, respectively, exceeds the baseline 1.59%, 1.84%, and 2.15% under the T.R. of 10%. For the SENet50, LHCBs on stage_2, $\text{stage}_{2,3}$, and $\text{stage}_{2,3,4}$, respectively, exceed the baseline 0.31%, 0.61%, and 1.09%. This demonstrates the effectiveness of the proposed LHCB. At the T.R. of 20%, for ResNet50, the improvements are, respectively, 1.32%, 1.58%, and 1.92% with LHCBs in different numbers of *stages*. For SENet50, the improvements are 0.74%, 1.16%, and 1.36%. The categories on which LHCB exceeds its baselines most are baseball diamond, tennis court, and basketball court, as shown in Fig. 6. From experimental results of datasets, AID and UC Merced, as shown in Fig. 5(b) and (c), it also can be found that as the number of LHCBs increases, the performance gradually improves. This is mainly because the introduction of high-frequency information enhances the network's ability of discriminating scenes with rich details. This further demonstrates the effectiveness of the proposed LHCB.

2) Impact on Different Downsample Methods: Fig. 7(a)–(c) shows the quantify comparison results of the effect of different downsample (pooling) methods in the network block LHCB with ResNet50 [33] and SENet50 [35] on datasets, NWPU-RESISC45, AID, and UC Merced, respectively. Two widely used downsample methods (MaxPool and MeanPool) are used to conduct experiments. From experimental results on dataset, NWPU-RESISC45, shown in Fig. 7(a), it can be found that the ResNet50 with MaxPool obtains the OA of 88.22%, and the ResNet50 with MeanPool obtains over accuracy with 89.32% at the T.R. of 10%. The difference is 1.10%. The SENet50 with MaxPool obtains the OA of 89.79%, and the SENet50 with MeanPool obtains the OA of 89.98% at the T.R. of 10%. The difference is 0.21%. The impact of pooling methods on the performance is relatively small under this T.R. At the T.R. of 20%, the ResNet50 with MaxPool obtains the OA of 92.18%, and the ResNet50 with MeanPool obtains over accuracy with 92.29%. This difference value is also small with 0.11%. Furthermore, the difference value is 0.07% for OA of SENet50 with MaxPool and the SENet50 with MeanPool. Combined with experimental results of datasets, AID and UC Merced, shown in Fig. 7(b) and (c), they illustrate that the downsample methods in LHCB have little effectiveness on the performance. Comparing with the results shown in Fig. 5, it can be found that the performance increase of LHCB is mainly brought by the high-frequency features generated by trainable Laplacian operator.

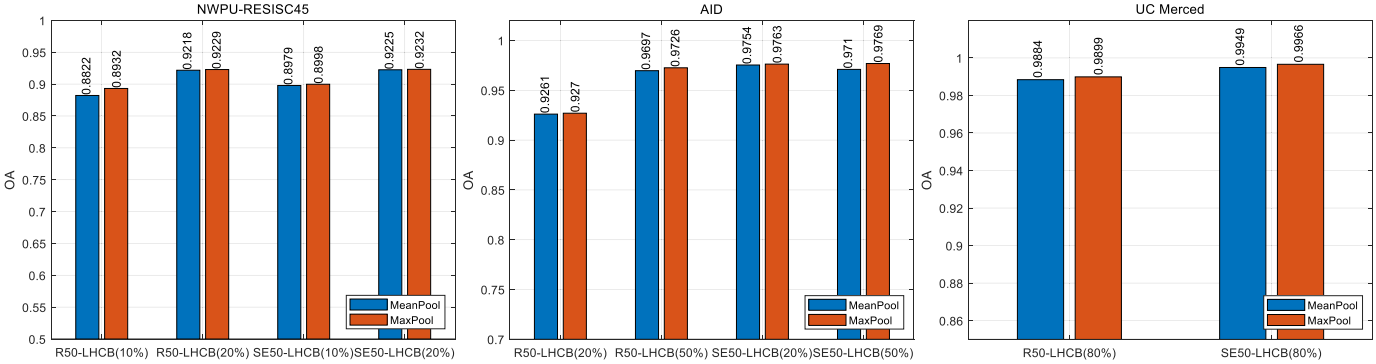
3) Impact on the Size of the Filter: Fig. 8(a)–(c), respectively, shows the quantify comparison results of the effect of integrating $\text{Conv}_{5\times5}$, $\text{Conv}_{3\times3}$, or not in LHCB with ResNet50 [33] and SENet50 [35] on datasets, NWPU-RESISC45, AID,

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(a) (b)

Fig. 6. Confusion matrixes generated by (a) SENet50 without LHCB and (b) SENet50 with LHCBs on NWPU-RESISC45 [1] under the T.R. of 20%.



(a) (b) (c)

Fig. 7. Effect of different downsample methods in LHCB with ResNet50 [33] and SENet50 [35] on (a) NWPU-RESISC45 [1], (b) AID [11], and (c) UC Merced [70].

and UC Merced. From experimental results on dataset, NWPU-RESISC45, shown in Fig. 8(a), the LHCB-ResNet50 without $\text{Conv}_{m\times n}$ obtains the OA of 88.31%, that with $\text{Conv}_{5\times 5}$ obtains over accuracy with 89.27%, and that with $\text{Conv}_{3\times 3}$ obtains the OA of 89.32% at the T.R. of 10%. In addition, the LHCB-SENet50 without $\text{Conv}_{m\times n}$ obtains the OA of 89.92%, that with $\text{Conv}_{5\times 5}$ obtains the OA of 89.95%, and that with $\text{Conv}_{3\times 3}$ obtains OA of 89.98%. It can be found that the performance without convolution operator is the worst, and when convolution is added, the size of the convolution has little effect on performance. At the T.R. of 20%, the same pattern can also be found. This demonstrates that the size of $\text{Conv}_{m\times n}$ in LHCB has little effectiveness of the performance on the dataset, NWPU-RESISC45. However, without $\text{Conv}_{m\times n}$ in LHCB results the lowest performance. Together with experimental results on datasets, AID and UC Merced, shown in Fig. 8(b) and (c), it can be found that the size of $\text{Conv}_{m\times n}$ in LHCB has little effect on classification performance. This is due to the fact that the trainable convolutional kernel could boost the fitting performance of the network.

D. Evaluation on Objective Function

To further verify the effectiveness of the objective function proposed for HFP, we conduct experiments on LH-ResNet and LH-SENet on dataset, NWPU-RESISC45. The experimental results are listed in Table III. It can be seen from Table III that the newly introduced objective function can improve the performance as a whole. This demonstrates the effectiveness of the designed objective function.

TABLE III

PERFORMANCE COMPARISON OF OBJECTIVE FUNCTION J ON NWPU-RESISC45 [1]

Methods	J_L	J
LH-ResNet50 (T.R. = 10%)	0.8901 ± 0.0030	**0.8932 ± 0.0021**
LH-SENet50 (T.R. = 10%)	0.8934 ± 0.0027	**0.8946 ± 0.0023**
LH-ResNet50 (T.R. = 20%)	0.9157 ± 0.0022	**0.9229 ± 0.0019**
LH-SENet50 (T.R. = 20%)	0.9203 ± 0.0025	**0.9251 ± 0.0012**
LH-ResNet101 (T.R. = 10%)	0.8911 ± 0.0018	**0.8969 ± 0.0012**
LH-SENet101 (T.R. = 10%)	0.8986 ± 0.0024	**0.9011 ± 0.0009**
LH-ResNet101 (T.R. = 20%)	0.9197 ± 0.0035	**0.9243 ± 0.0010**
LH-SENet101 (T.R. = 20%)	0.9215 ± 0.0033	**0.9261 ± 0.0017**

E. Evaluation on HFP

To verify the generalization performance of the proposed HFP generated by LHCB, we embed HFPs into several existing CNN structures for experiments. We choose the popular SE CA module improved SENet [35], convolutional channel (CA), and SA block improved (CBAM) [37] as the compared methods. We conduct experiments on the dataset, NWPU-RESISC45, since it has the most samples and categories than other two datasets. The basic network selected is ResNet101 [33].

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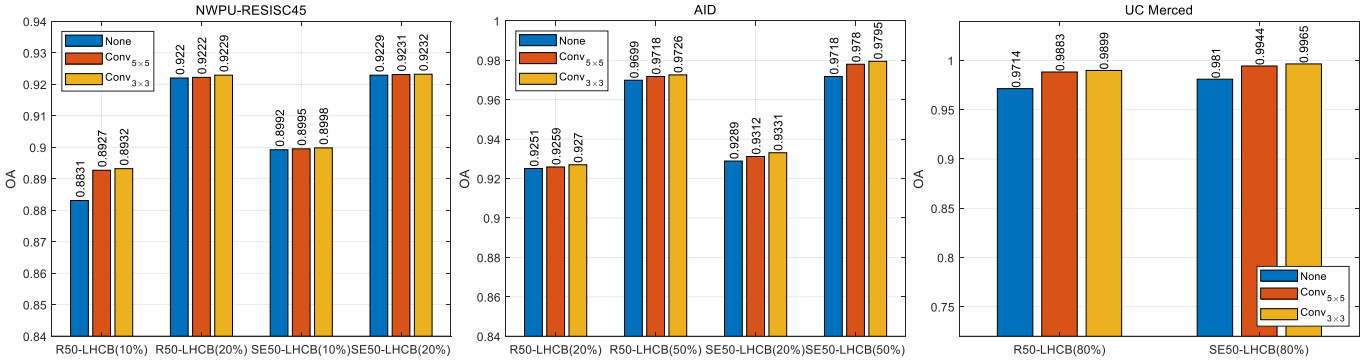


Fig. 8. Effect of integrating different sizes of $\text{Conv}_{m\times n}$ in LHCB with ResNet50 and SENet50 on (a) NWPU-RESISC45 [1], (b) AID [11], and (c) UC Merced [70].

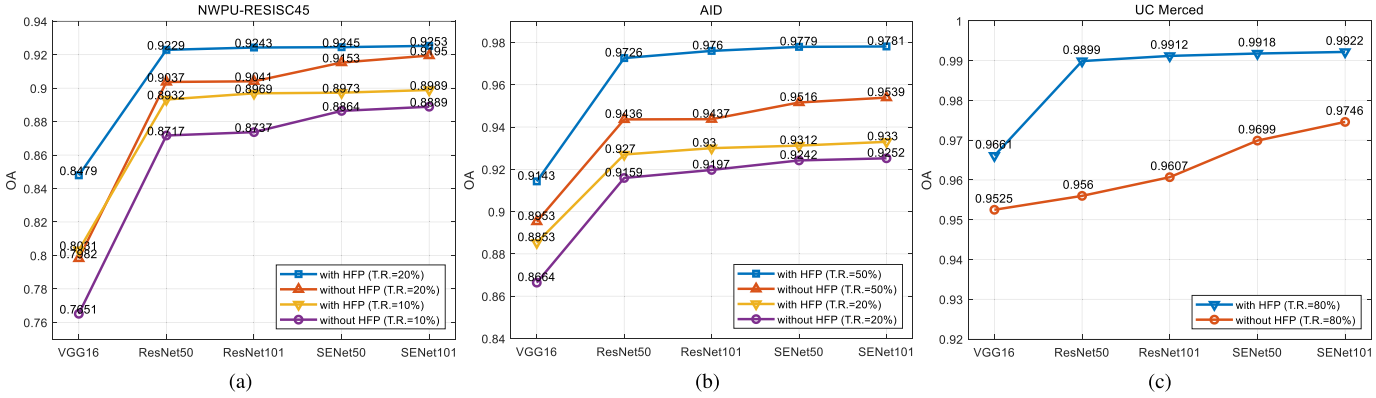


Fig. 9. CNNs with or without HFP. (a) NWPU-RESISC45. (b) AID. (c) UC Merced.

TABLE IV
PERFORMANCE COMPARISON OF LHCB WITH SEVERAL IMPROVED CNN MODULES ON NWPU-RESISC45 [1]

Methods	T.R. = 10%	T.R. = 20%
ResNet101 [33]	0.8737 ± 0.0021	0.9041 ± 0.0023
ResNet101 + LHCB	**0.8969 ± 0.0012**	**0.9243 ± 0.0010**
ResNet101 + CA_{SE} [35]	0.8889 ± 0.0010	0.9195 ± 0.0043
ResNet101 + CA_{SE} + LHCB	**0.8989 ± 0.0012**	**0.9253 ± 0.0010**
ResNet101 + CA_{CBAM} [37]	0.8852 ± 0.0012	0.9128 ± 0.0020
ResNet101 + CA_{CBAM} + LHCB	**0.8991 ± 0.0009**	**0.9253 ± 0.0018**
ResNet101 + SA_{CBAM} [37]	0.8854 ± 0.0015	0.9132 ± 0.0010
ResNet101 + SA_{CBAM} + LHCB	**0.8982 ± 0.0022**	**0.9260 ± 0.0014**
ResNet101 + CA_{CBAM} + SA_{CBAM}	0.8887 ± 0.0011	0.9188 ± 0.0030
ResNet101 + CA_{CBAM} + SA_{CBAM} + LHCB	**0.9001 ± 0.0021**	**0.9269 ± 0.0020**

From Table IV, at the T.R. of 10%, the LH-ResNet101 (ResNet101 + LHCB) has the OA with 89.69%, which is higher than the baseline (ResNet101) with 2.32%, the LH-SENet101 (ResNet101 + CA_{SE} + LHCB) has the OA with 89.89%, which is higher than the baseline SENet101 (ResNet101 + CA_{SE}) with 1.00%, the ResNet101 + CA_{CBAM} + LHCB has the OA with 89.91%, which is higher than the baseline with 1.39%, the ResNet101 + SA_{CBAM} + LHCB has the OA with 89.82%, which is higher than the baseline with 1.28%, and the performance of ResNet101 + CA_{CBAM} + SA_{CBAM} + LHCB is higher than the baseline with 1.14%. At the T.R. of 20%, it also can be found that network with LHCB could further improve the performance.

Combined with the experimental results at the T.R. of 10% and 20%, the performance of the proposed LHCB is better than the SE attention module and the CBAM attention module on dataset, NWPU-RESISC45. This is mainly because the introduced high-frequency features can assist the feature representation of scenes with rich texture features and, therefore, promote to correctly classify image scenes. In addition, this indicates that the proposed LHNet can be designed based on the existing CNN structures to further improve the performance.

F. Comparison With Popular and State-of-the-Art Approaches

To put the classification performance into perspective, we compare two types LHNets (based on ResNet50 and ResNet101, respectively) to several classical and state-of-the-art image scene classification approaches on three public available datasets, NWPU-RESISC45 [1], AID [11], and UC Merced [70]. The classical handcrafted feature-based approaches selected are the LBP [21], color histogram (CH [20]), and GIST [22]. The classical unsupervised learnable feature-based approaches selected are spatial pyramid matching (SPM) kernel [30], (BoVWs [29]), and locality-constrained linear coding (LLC [31]). The CNN-based approaches selected are the popular VGG-16 [32], GoogLeNet [34], ResNet (ResNet50 and ResNet101) [33], and SENet (SENet50 and SENet101) [35]. The classical remote sensing image scene classification approaches selected are recently proposed

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TABLE V
COMPARISON WITH POPULAR AND STATE-OF-THE-ART IMAGE SCENE CLASSIFICATION APPROACHES

Methods	Publication Year	Backbones	UC Mercerd [70]	AID [11]		NWPU-RESISC45 [1]	
			T.R. = 80%	T.R. = 20%	T.R. = 50%	T.R. = 10%	T.R. = 20%
LBP [21]	TPAMI2002	-	0.3619 ± 0.0170	0.2621 ± 0.0043	0.2976 ± 0.0035	0.1926 ± 0.0034	0.2151 ± 0.0019
CH [20]	IJCV1991	-	0.4611 ± 0.0109	0.3415 ± 0.0030	0.3736 ± 0.0042	0.2472 ± 0.0012	0.2759 ± 0.0009
GIST [22]	IJCV2001	-	0.4670 ± 0.0119	0.3042 ± 0.0045	0.3512 ± 0.0031	0.1583 ± 0.0021	0.1781 ± 0.0020
SPM (SIFT) [30]	CVPR2006	-	0.6135 ± 0.0245	0.3815 ± 0.0067	0.4525 ± 0.0069	0.2772 ± 0.0059	0.3301 ± 0.0045
BoVW (SIFT) [29]	ICCV2003	-	0.7409 ± 0.0310	0.6132 ± 0.0034	0.6755 ± 0.0042	0.4176 ± 0.0019	0.4493 ± 0.0021
LLC (SIFT) [31]	CVPR2010	-	0.7127 ± 0.0200	0.5631 ± 0.0063	0.5997 ± 0.0053	0.3891 ± 0.0025	0.4013 ± 0.0037
VGG-16 [32]	ICLR2015	VGG-16	0.9525 ± 0.0123	0.8664 ± 0.0027	0.8953 ± 0.0032	0.7651 ± 0.0017	0.7982 ± 0.0013
GoogLeNet [34]	CVPR2015	GoogLeNet	0.9429 ± 0.0074	0.8347 ± 0.0035	0.8643 ± 0.0065	0.7623 ± 0.0032	0.7851 ± 0.0021
ResNet50 [33]	CVPR2016	ResNet50	0.9560 ± 0.0064	0.9159 ± 0.0020	0.9436 ± 0.0017	0.8717 ± 0.0033	0.9037 ± 0.0019
ResNet101 [33]	CVPR2016	ResNet101	0.9607 ± 0.0068	0.9197 ± 0.0028	0.9437 ± 0.0038	0.8737 ± 0.0021	0.9041 ± 0.0023
SENet50 [35]	CVPR2018	SENet50	0.9699 ± 0.0112	0.9242 ± 0.0030	0.9516 ± 0.0017	0.8864 ± 0.0027	0.9153 ± 0.0023
SENet101 [35]	CVPR2018	SENet101	0.9746 ± 0.0051	0.9252 ± 0.0027	0.9539 ± 0.0029	0.8889 ± 0.0010	0.9195 ± 0.0043
D-CNN [14]	TGRS2018	VGG16	0.9895 ± 0.0012	0.9078 ± 0.0014	0.9691 ± 0.0120	0.8921 ± 0.0057	0.9183 ± 0.0025
D-CNN [14]	TGRS2018	GoogLeNet	0.9706 ± 0.0011	0.8873 ± 0.0012	0.9632 ± 0.0011	0.8684 ± 0.0013	0.9045 ± 0.0014
MG-CAP [49]	TIP2020	VGG-D	0.9860 ± 0.0026	0.9211 ± 0.0015	0.9514 ± 0.0012	0.8942 ± 0.0019	0.9172 ± 0.,0016
SCCov [51]	TNNLS2020	Alexnet	0.9804 ± 0.0023	0.9110 ± 0.0015	0.9330 ± 0.0013	0.8433 ± 0.0026	0.8730 ± 0.0023
SCCov [51]	TNNLS2020	VGG16	0.9905 ± 0.0025	0.9312 ± 0.0025	0.9610 ± 0.0016	0.8930 ± 0.0035	0.9210 ± 0.0025
Adv.Tr. [50]	TGRS2021	ResNet152	0.9710 ± 0.0046	0.8514 ± 0.0084	0.9104 ± 0.0050	0.8189 ± 0.0020	0.8653 ± 0.0042
Adv.Tr. [50]	TGRS2021	ResNeXt101	0.9790 ± 0.0059	0.8693 ± 0.0109	0.9244 ± 0.0030	0.8316 ± 0.0025	0.8762 ± 0.0034
PSGAN [3]	TGRS2022	ResNeXt101	0.9848 ± 0.0064	0.8947 ± 0.0034	0.9267 ± 0.0055	0.8472 ± 0.0072	0.8847 ± 0.0056
PSGAN [3]	TGRS2022	SCNet	0.9643 ± 0.0168	0.8718 ± 0.0065	0.8853 ± 0.0300	0.8847 ± 0.0056	0.8864 ± 0.0077
LHNet	Ours	ResNet101	0.9912 ± 0.0010	0.9300 ± 0.0008	0.9760 ± 0.0008	0.8969 ± 0.0012	0.9243 ± 0.0010
LHNet	Ours	SENet101	**0.9922 ± 0.0011**	**0.9330 ± 0.0010**	**0.9781 ± 0.0013**	**0.8989 ± 0.0015**	**0.9253 ± 0.0013**

D-CNN [14], MG-CAP [49], Adv.Tr. [50], PSGAN [3], and SCCov [51]. For fair comparison, the parameter settings in compared methods keep the settings in the corresponding papers.

Table V shows the comparison results. In general, the performance of CNN-based methods is better than the performance of unsupervised learnable feature-based approaches, and the performance of unsupervised learnable feature-based approaches is better than the performance of methods based on manual features on different datasets and T.R. In addition, all the LHNets based on CNN networks outperform the corresponding baselines, as shown in Fig. 9. The proposed method achieves the best performance on datasets, AID, UC Merced, and NWPU-RESISC45. This demonstrates the state-of-the-art performance of the proposed approaches. This is mainly due to the superiority of the high-frequency features introduced to increase the discriminative ability of LHNet and, therefore, increase the performance. In addition, the proposed approaches are more stable than most compared approaches from the standard deviation. This further demonstrates the superiority of the proposed method.

V. CONCLUSION AND FUTURE WORK

In this article, we propose a novel deep texture D-CNN (LHNet), which aims at improving the discriminate ability of classifying the remote sensing image scenes with obvious texture features. Such texture features are especially important to identify objects. Traditional CNNs with hierarchical filtering and downsampling operators can be taken as low-frequency pathway. Based on CNN, we introduce an HFP that contains high-frequency feature maps generated by trainable Laplacian operator in each layer. The feature maps in low-frequency pathway and HFP interact with each other during the propagation process. To increase the intraclass similarity of high-frequency features, we introduce a new objective for HFP. We choose the most popular ResNet and SENet as the backbone networks to conduct experiments. A large number of experiments demonstrate the superior performance of the proposed method.

In the future, we will integrate other multiscale analysis architectures into deep networks to learn more discriminative features for remote sensing image scene classification.

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